Self-supervised learning applied to speaker and language recognition

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Machine learning and supervised training

Question: Why are large human-labeled datasets important?

- Recent models tend to increase the depth of convolutional layers to improve their capacity.
- Adding more trainable parameters may lead to overfitting (unbalanced bias-variance tradeoff).
- In order to compensate this, using a larger training set is mandatory.

The limitations of supervised and unsupervised learning

Supervised

Learning a function that maps an input to an output based on human-labeled input-output pairs.

Unsupervised

Learning a function mapping an input to an output based only on the input (unlabeled data).

Drawbacks of supervised learning:

- Labeling datasets is expensive, tedious and slow.
- Manual labeling is not scalable to the amount of data available today.
- It could lead to biased models towards the considered problem.

Introducing self-supervised learning

Self-supervised

Similar to supervised learning but with automatically generated targets. The principle is to define a **pretext task** to learn useful representations for a **downstream task** (through transfer learning for instance).

Example of some **pretext tasks**:

- Predicting the relative position of two sub-regions of the input image[1].
- Predicting parts of the input that has been removed[2].
- Predicting spatial transformations applied to the input[3].

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[1] Carl Doersch/Abhinav Gupta/Alexei A. Efros: Unsupervised Visual Representation Learning by Context Prediction, 2016.

[2] Richard Zhang/Phillip Isola/Alexei A. Efros: Colorful Image Colorization, 2016.

[3] Spyros Gidaris/Praveer Singh/Nikos Komodakis: Unsupervised Representation Learning by Predicting Image Rotations, 2018.

An example for computer vision: image colorization

Learning how to colorize images implies learning meaningful semantic information from images[4].

- **Pretext task:** predict color channel values given a grayscale image.
- **Downstream task:** image classification on ImageNet (achieves state-of-the-art accuracy).

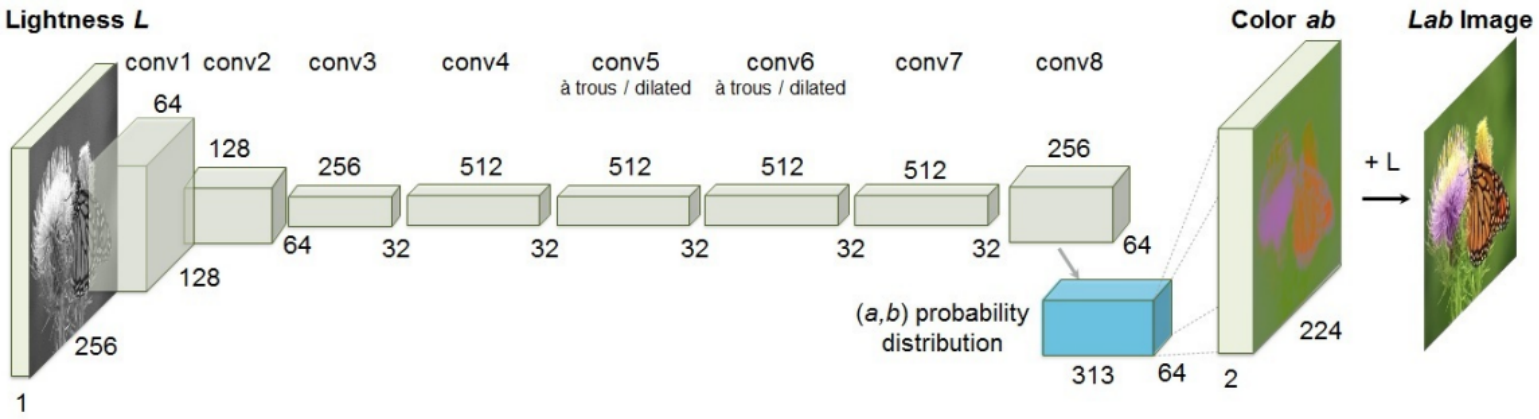

Figure: Model architecture used to perform image colorization.

[4] Zhang/Isola/Efros: Colorful Image Colorization (see n. 2).

An architecture for speech signals: PASE

PASE[5] outputs speech representations from raw audio.

- First layer (SincNet[6]) acts as a band pass filter.
- 7 x regular 1D convolutional layers.
- Outputs a feature vector of length 100.

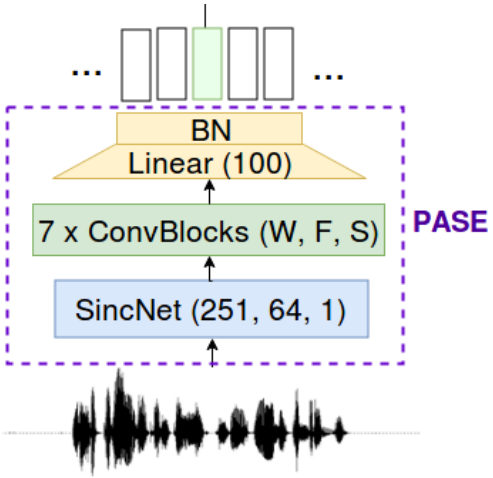

Figure: PASE architecture.

[5] Santiago Pascual et al.: Learning Problem-agnostic Speech Representations from Multiple Self-supervised Tasks, 2019.

[6] Mirco Ravanelli/Yoshua Bengio: Speaker Recognition from Raw Waveform with SincNet, 2019.

Self-supervised training of PASE

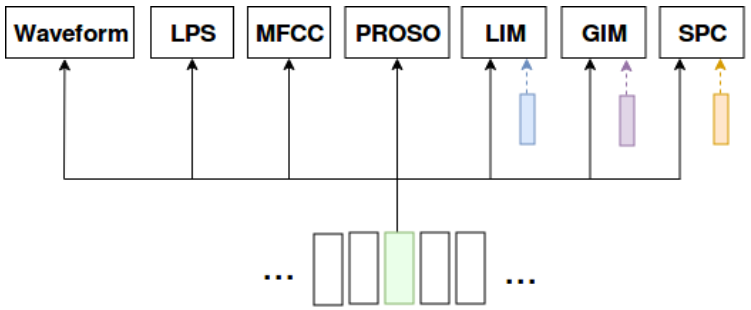

Figure: Set of PASE self-supervised workers.

PASE is trained jointly with 7 self-supervised modules:

- **Regressors:** extracting common speech features (MFCC, LPS).
- **Discriminators:** maximizing mutual information between representations of speech sampled from the same sentence (Local Info Max[7]).

[7] Mirco Ravanelli/Yoshua Bengio: Learning Speaker Representations with Mutual Information, 2019.

Objective of my work

Problem:
PASE is problem-agnostic which implies that it was not specifically made for speaker and language recognition.

Objective:
Design an efficient speaker and language recognition model based on the main concepts introduced by PASE architecture.

Current progress and perspectives

Work achieved during the month:

- Read and summarize ten research articles related to the field.
- Conduct experiments with PASE and pre-trained models to solve the speaker recognition task.

Task planned for this month:

- Start by re-implementing PASE encoder in Keras in order to reproduce the results obtained in the original publication.

Questions?

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References

- Doersch, Carl, Abhinav Gupta, and Alexei A. Efros: Unsupervised Visual Representation Learning by Context Prediction, 2016.
- Gidaris, Spyros, Praveer Singh, and Nikos Komodakis: Unsupervised Representation Learning by Predicting Image Rotations, 2018.
- Pascual, Santiago et al.: Learning Problem-agnostic Speech Representations from Multiple Self-supervised Tasks, 2019.
- Ravanelli, Mirco and Yoshua Bengio: Learning Speaker Representations with Mutual Information, 2019.
- Idem: Speaker Recognition from Raw Waveform with SincNet, 2019.
- Zhang, Richard, Phillip Isola, and Alexei A. Efros: Colorful Image Colorization, 2016.